

Hybrid Neural Pipelines for TSP Problem: When Diffusion Meets Learned Local Search

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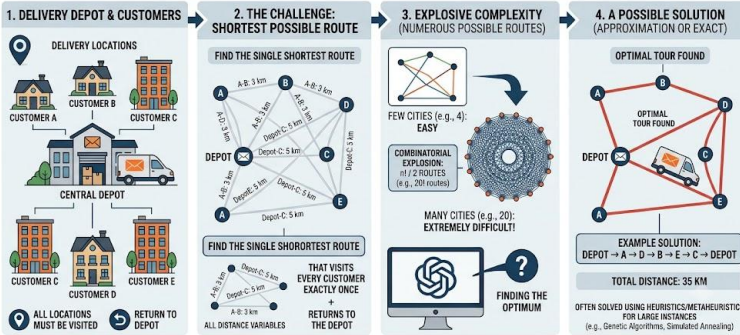
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Background & Motivation

The **Traveling Salesman Problem (TSP)** asks: given n locations, find the **shortest Hamiltonian cycle** visiting each exactly once. Despite its simple formulation, TSP is **NP-hard** and sits at the **heart of modern logistics**: from Meituan's food delivery routing to Taobao's warehouse optimization, from PCB drilling to genome sequencing.

THE TRAVELING SALESPERSON PROBLEM (TSP): PACKAGE DELIVERY EDITION



TSP is algorithmically unique because it:

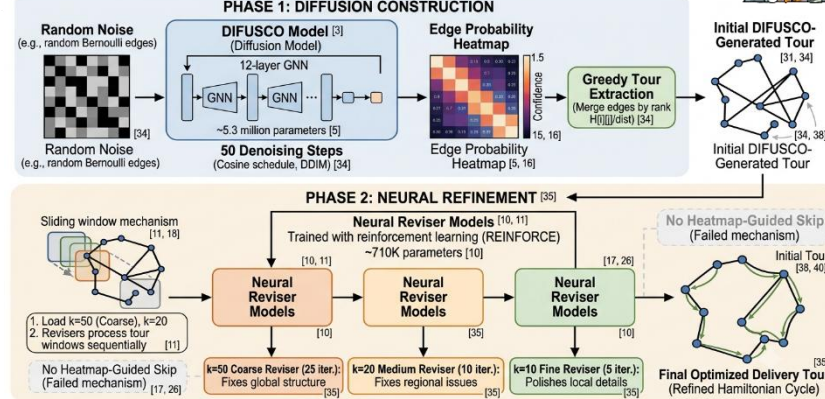
- Is **NP-hard**, yet admits a $1.5\times$ constant-factor approximation (Christofides)
- Is **simple enough** for deep learning to **learn useful heuristics**
- Is **complex enough** that **pure ML fails** without classical components
- Serves as the **canonical benchmark** for new algorithmic ideas

Algorithm Landscape

- Nearest Neighbor**—**Greedy baseline**: repeatedly visits closest unvisited node
- Christofides**—The only **polynomial**: MST $\rightarrow$ Blossom MWPM $\rightarrow$ Eulerian shortcut
- C+2opt**—**Christofides followed by 2-opt** local search that eliminates crossing edge pairs
- LKH3**—Generalizes **k-opt** moves guided by α -nearness from minimum 1-trees

Method	Paradigm	Complexity	Key Property
Nearest Neighbor	Greedy heuristic	$O(n^2)$	20–30% gap (floor)
Christofides (1976)	Approximation	$O(n^3)$	$1.5\times$ guarantee
C+2opt	Improvement	$O(n^3)$	$\sim 4.3\%$ gap (baseline)
LKH3 (Helsgaun 2017)	Local search	$\sim O(n^{2.2})$	0.4% gap (ceiling)
DIFUSCO (NeurIPS'23)	Diffusion model	$O(n^2 d^2)$	Generative, 5.3M params
DualOpt (AAAI'25)	Divide + RL	$O(nk^2)$	Improvement, 710K params

Difusco-→DualOPT Hybrid Pipeline



Core contribution — taking the **generative power of diffusion** from DIFUSCO and the **learned local search** from DualOpt. The hybrid pipeline lets each component do what it does best.

- DIFUSCO** ^[1] (NeurIPS 2023) It's diffusion process captures **GLOBAL tour structure** — the GNN sees the entire $n\times n$ adjacency matrix simultaneously. But its raw output has local defects (self-crossings, suboptimal turns). Casts TSP as categorical denoising diffusion over graph adjacency matrices.
- DualOpt** ^[2] (AAAI 2025) Dual divide-and-optimize strategy — grid-based plane partitioning + LKH3 sub-solves, then cascaded neural revisers at **LOCAL refinement** — trained via **REINFORCE** to directly minimize tour length within sliding windows. But they need a reasonable starting tour.

Experimental Results

We evaluate on three datasets: (1) **Random Uniform TSP** — 10 held-out instances at each of $n = 50, 100, 200, 500, 1000$, matching DIFUSCO's training distribution; (2) **TSPLIB Standard Benchmark** — 8 symmetric instances (51 to 1002 nodes) with known optimal values, the authoritative test of generalization; (3) **Clustered Delivery TSP** — 4 Gaussian clusters + scattered points at $n = 50, 100, 200$, representing real-world delivery density patterns. All methods evaluated on identical instances.

DIFUSCO+2opt
Cost: 12.49

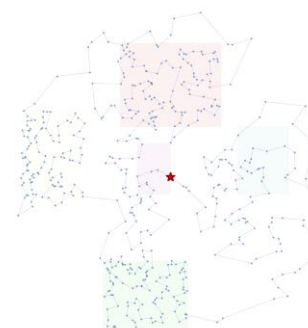


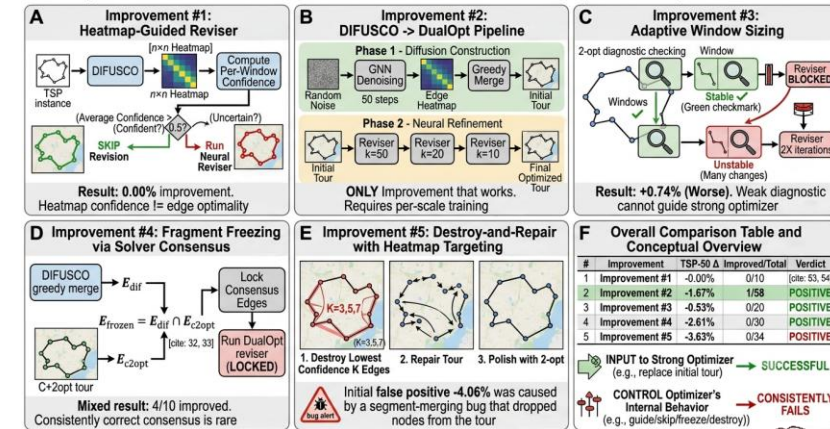
Table 4: DIFUSCO $\rightarrow$ DualOpt Hybrid Pipeline (TSP-50, 10 instances).

Method	Mean Cost	vs GT	vs Raw
DIFUSCO raw	6.496	+13.67% baseline	
DIFUSCO + 2-opt	5.969	+1.32% $\sim 10.87\%$	
DIFUSCO $\rightarrow$ DualOpt (ours)	5.793	-1.67% $\sim 13.50\%$	
C+2opt (baseline)	5.891	baseline $\sim 12.03\%$	

Table 2: TSPLIB Standard Benchmark — Generalization Across Scales.

Instance	n	NN	C+2opt	DIFUSCO	DualOpt
eil51	51	20.6%	3.7%	5.4%	1.0%
berlin52	52	19.1%	4.6%	13.2%	0.03%
eil76	76	32.3%	7.3%	6.9%	3.8%
kroA100	100	26.2%	5.8%	10.0%	3.8%
ch150	150	25.5%	3.4%	4.6%	45.3%
ts225	225	23.3%	3.1%	3.5%	36.8%
a280	280	22.1%	2.8%	7.7%	6.7%
pr1002	1002	21.8%	3.9%	7.1%	16.4%
Mean		23.9%	4.3%	6.9%	2.2% [†]

Systematic Architecture and Performance of Neural TSP Optimizer Improvements



Key Insights

- HYBRID EITHER ALONE.** DIFUSCO (global) + DualOpt (local) = -1.67% vs GT.
- CHANGE THE INPUT, NOT THE PROCESS.** Replacing the initial tour (#2) **succeeded**; guiding the reviser internally (#1,3,4,5) **failed** — it already saturates.
- CLASSICAL WINS AT SMALL SCALES.** At $1e1000$, LKH3 (0.4% gap, 1.2s) and C+2opt are fast, near-optimal, and **need no training**.
- NEURAL METHODS TARGET LARGE SCALES.** DualOpt beats LKH3 by 1.40% on TSP-100K (104 times faster); DIFUSCO scales to TSP-10K. Our ceiling was $n=1000$ on a single RTX 2060 — the hybrid pipeline should be even more impactful at scale.
- FIVE IMPROVEMENTS CHART THE DESIGN SPACE.** #1: confidence $\neq$ guidance. #2: hybrid pipelines work. #3: weak optimizers can't guide strong ones. #4: solver consensus shows promise (4/10 improved, up to -5.16%) but needs a tiebreaker. #5: once saturated perturbations only degrade.

References

- [1] Sun & Yang. DIFUSCO: Graph-based Diffusion Solvers for Combinatorial Optimization. NeurIPS, 2023.
- [2] Zhou et al. DualOpt: A Dual Divide-and-Optimize Algorithm for the Large-Scale Traveling Salesman Problem. AAAI, 2025.
- [3] Christofides. Worst-case analysis of a new heuristic for the travelling salesman problem. Technical Report, 1976.
- [4] Helsgaun. An Extension of the Lin-Kernighan-Helsgaun TSP Solver. 2017.
- [5] Kool, van Hoof & Welling. Attention, Learn to Solve Routing Problems! ICLR, 2019.